

Does Urban Sprawl Cause Residential Vacancy Across French Municipalities?

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Abstract

This paper studies residential vacancy as a by-product of housing construction, which could mitigate the effect of supply-side policies on housing prices. We estimate the causal effect of urban sprawl, measured as the amount of newly developed land per inhabitant, on vacancy rates in French municipalities. Since urban sprawl is endogenous, we use slope and soil stability as instrumental variables that are assumed to affect vacancy rates only through differentiated sprawl capacities. From 1995 to 2019, we find that median annual sprawl (+27 m²/inhab.) increases the average vacancy rate by +0.33 percentage points, equivalent to +2.85 vacant dwellings per municipality. This result passes the over-identification test and is robust to spatial spillovers accounted for by data aggregation. Heterogeneity analysis shows that the estimated causal effect is more pronounced for short-term vacancies, for apartments rather than houses, for older dwellings, and in areas with declining employment or higher initial vacancy rates. Our results highlight the importance of taking into account households' differentiated preferences for dwelling and neighbourhood characteristics in order to design more relevant public policies to increase housing affordability.

Keywords: affordability crisis • vacant housings • land use-regulation • urban expansion • real estate • instrumental variable

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1 Introduction

In many countries, housing price growth outpaced wage growth in the last decades, which increased the share of the household budget spent on housing (Wetzstein, 2017). In France, this share increased from 23.2% in 1995 to 28.4% in 2020, and housing is now the largest item of household expenditure. As a result, housing affordability is becoming a major concern for citizens and public policy (Gabriel and Painter, 2020), with many households having to make tougher trade-offs with consumption and life choices (Newman and Holupka, 2016).

The economic literature suggests that one way out of this crisis is to increase the supply of housing through new construction (Saiz, 2023) and calls for a relaxation of land regulation (Cheshire, Hilber, and Koster, 2018). Indeed, it is likely that favouring housing supply reduce average house prices, although the extent of the expected response is debatable (Freemark, 2024). In particular, housing vacancy is a recurrent and significant phenomenon that could mitigate the downward price response expected from new construction.¹ According to the French National Statistics Institute (INSEE), the number of vacant dwellings has increased by an average of 1.3% per year since 1980.² More specifically, between 1983 and 2000, there were around 3.4 empty dwellings per 100 inhabitants, whereas this ratio is by 4.6 points higher in 2020. Thus, the French housing stock presents about 10% of vacant dwellings in the recent years.

For a closed city with an inelastic housing demand D , Figure 1 illustrates the expected effect of a rightward shift in the supply function from S to S' (e.g. from the decrease in land prices due to the relaxation of land regulation). For homogeneous dwellings, the supply shift classically moves the market equilibrium to the point B with a lower dwelling price $P' < P$. If, instead, we consider that new dwellings are preferred to old dwellings at the initial price P ,³ the resulting equilibrium is A . In this latter case, the effect of the supply shift on the housing price is zero, and each new dwelling *causes* an additional vacancy in the old housing stock. From these two polar cases in terms of household preferences for new dwellings, we expect the resulting equilibrium to lie somewhere on the $A - B$ segment. The vacancy caused by the supply shift would be posi-

¹ To save space, the theory of housing vacancy is presented in Annex A.1 of the Online Annex (OA).

² <https://www.insee.fr/en/statistiques/5887128>.

³ As incomes rise, individuals desire larger garden areas along with larger living spaces, while advances in transport facilities have made it easier for individuals to reach remote locations (Brueckner, 1980).

tive, but less than $(Q' - Q)$, and the price fall would be less than $(P - P')$. Taking into account that new dwellings are imperfectly substitutes to the old ones, [Figure 1](#) shows that housing vacancy is a by-product of housing construction. From a statistical view, housing vacancy is a mediating variable that reduces the expected downward effect of the supply shift on housing prices.

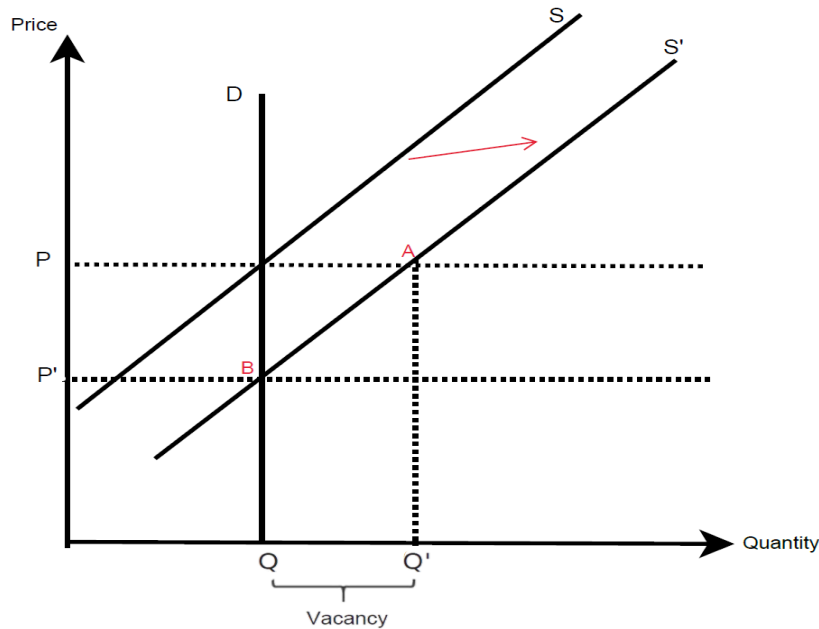


Figure 1: Housing market outcome of a supply shift with residential vacancy.
Source: Authors' realisation.

Empirically, we study the effect of urban sprawl as a shift in housing supply that is expected to generate vacant dwellings. In line with its usual definition as "the tendency towards lower urban densities as the urban area expands" ([Nechyba and Walsh, 2002](#), p.178), we compute urban sprawl at the municipal level as the ratio of newly developed land to population size. Rather than using a composite index that is not directly related to the supply of new dwellings and household preferences ([Galster et al., 2001](#)), we choose this measure to account for density differences between old and new dwellings and for heterogeneous population size across municipalities.⁴ It is well recognised that housing supply is mainly constrained by the cost of building on the urban fringe ([Saiz, 2010](#)) and that households have strong preferences for low densities ([Glaeser and Kahn, 2004](#)). These findings make urban sprawl a relevant supply-side variable for studying the effect of housing construction on housing vacancy.

⁴ Urban sprawl is an incomplete definition of housing supply, as it is possible for supply-side policies to increase the housing stock by increasing density for a given urban footprint.

In order to estimate a causal effect, we take into account open city effects that are not considered in the stylised model above. All else equal, more 'attractive' municipalities with more dynamic labour markets or more positive amenities have both higher urban sprawl and lower residential vacancy rates. According to the omitted bias formula, a raw OLS regression would underestimate the causal effect of interest. Our identification strategy is then to regress vacancy rates on urban sprawl predicted by instrumental variables that affect urban sprawl independently of 'attractiveness'. Physical variables about sprawling capacity cannot change in response to changes in housing demand, so they are unlikely to be correlated with vacancy rates other than through urban sprawl.⁵ Hence, we propose to identify the causal effect of urban sprawl on vacancy rates from variations in the supply function predicted from cross-sectional variations in land slope and soil stability (Saiz, 2010; Hilber and Vermeulen, 2016; Ay et al., 2018). Through construction costs, high slope and low soil stability are expected to have a negative impact on urban sprawl across municipalities, which is confirmed by the data.

Due to varying empirical methods, there is no consensus in the literature on the effect of urban sprawl on residential vacancy. On the right hand, the raw correlation analysis of Bowman and Pagano (2000) shows that US cities that experienced both spatial expansion and population growth have more residential vacancy than cities with a fixed footprint. Still for the US, Smirnova and Ingalls (2007) illustrates how low levels of urban sprawl provide greater opportunities for growth in city centres, thus reducing the likelihood of increased vacancy rates. Xiao et al. (2006) obtain similar results in China, showing that cities with higher levels of sprawl and vacancy tend to have more attractive local economies and higher tax bases, and are less inclined to encourage infill development. Newman et al. (2016) analyse the top 20 elastic (land area expanded the most) and inelastic (land area contracted the most) US cities with over 100,000 inhabitants from 2000 to 2010. They find from model with fixed effects that sprawl contributes significantly to vacancy by increasing the housing supply, which is interpreted as reducing the cost of housing and thus the opportunity cost of maintaining residential vacancy for homeowners.

On the other hand, some studies show that urban sprawl contributes to the reduction of res-

⁵ A more complete assessment of the effect of housing supply on the affordability crisis would be to examine house prices (e.g. Hilber and Vermeulen, 2016). However, most of these analyses account for endogeneity through market clearing conditions (where supply equals demand), and identifying the impact of excess supply (vacancies) on housing prices would require a different empirical strategy than the one we propose in this paper.

idential vacancy, directly or indirectly. [Rusk \(2003\)](#) finds that increasing the urban footprint has a direct effect of decreasing vacancy by strengthening the central city. This allows to limit urban decline and reduce the number of vacant properties that are mainly located at the center. Using data for 350 English local authorities between 1981 and 2011, [Cheshire, Hilber, and Koster \(2018\)](#) investigate how land-use regulation affects vacancy rates. Their empirical methodology involves an instrumental variable approach that accounts for the "attractiveness" confounding problem, using variables about the political composition of local authorities and local vote shares to instrument land-use regulation. Their results suggest that tighter local planning restrictions about land use lead to higher local vacancy rates, which indirectly indicate a negative effect of potential urban sprawl on residential vacancy. This is explained by a "mismatch effect" between supply and demand, which contradicts the intuitions from [Figure 1](#).

This paper contributes to the literature by proposing a new identification strategy and by using France as a case study with a pseudo-panel of $N \approx 35,000$ municipalities from 1995 to 2019. Our results indicate that urban sprawl has a statistically significant positive causal effect on residential vacancy, which is shown to be rather far from main city centers. A 10 m^2 increase in developed land per inhabitant increases the vacancy rate by 0.13 percentage points, which corresponds to an average of 2.5 housing units per municipality. This effect remains of the same order of magnitude while controlling for average dwelling age, comfort, and size, for given average household size, age, and income, and for given location along the urban-rural gradient. It is worth noting that the effect is more pronounced for flats rather than houses, for short-term vacancy rather than for long-term, for economically disadvantaged areas with declining employment, for initially high vacancy rates, and for municipalities further from cities. Finally, we find that this effect is amplified by some control variables about dwellings (age, comfort), households (size), and the location of municipalities, which indicates the importance of a good match between household preferences and supplied dwellings' characteristics.

Our results contribute to a better understanding of the impact of urban sprawl on residential vacancy, highlighting the heterogeneity of the impact according to varying economic, demographic, and geographical factors. These findings are particularly relevant for urban and real estate development policies aimed at simultaneously reducing residential vacancy and increas-

ing housing affordability. Our study highlights that the observation of high average housing prices is not only a result of housing shortages, but could also arise from the interaction between the quality of existing dwellings and the preferences of households. However, our results do not contradict the possibility that an increase in housing supply would reduce house prices, but they do suggest that this effect may be smaller than expected from standard market clearing models. This has strong policy implications because, from a welfare perspective, the benefits of new construction (that are generally located further from city centres as in our sprawl measure) need to be balanced against the associated infrastructure and environmental costs.

The rest of the paper is structured as follows. Section 2 presents the data and provides descriptive statistics. It is followed by Section 3 that outlines our empirical strategy. Section 4 presents our results, and reports the additional findings from robustness and heterogeneity analyses. Section 5 concludes by discussing the policy implications of our results.

2 Data

We gather data for all French municipalities ($N \approx 35,000$) between 1995 and 2019. For administrative reasons, our sample is restricted to odd-numbered years in addition to 2010.

2.1 Sources

Housing taxes (FILOCOM). A comprehensive database on residential vacancy in France comes from FILOCOM (*Fichiers des Logements à la COMMune*, Segú, 2020),⁶ an official source produced by the general direction of public finances for the administration of the ministry of housing. These data report about 30 million housing units per year with the municipalities to which they belong. They primarily contain the type of housing occupation with a vacancy category and the length of the current vacancy period if relevant. Data also have numerous auxiliary

⁶ There are three measures of residential vacancy available for France. The first is computed by INSEE from an extensive definition (that groups dwellings available for sale or rent, or held by firms for future use by employees) assessed at the census dates (<https://www.insee.fr/fr/metadonnees/definition/c1059>). The second available measure comes from the electricity supplier company (*Électricité de France*) with a definition based on annual electricity consumption. The third source is the one used in this paper, described in more detail in the text.

variables about dwelling attributes (e.g. construction date, living space, comfort level, building traits) and household characteristics (e.g. income, household size, occupant and owner ages). Our econometric analysis is based on the aggregation of these variables at the municipality level.

Building permits (SITADEL). A second data source is SITADEL (*Système d'Information et de Traitement Automatisé des Données Élémentaires sur les Logements*) that contains all allowed building permits in France since 1972. They are recorded at the approval date and are located at the municipal level. This source informs the amount of land allowed to construct new housing units, both by infilling city footprint and at the urban fringe. These data have the advantage of distinguishing commercial and residential land development (Ay et al., 2018), from which we compute the land areas of approved residential permits. These areas include flats, grouped and detached houses that are allowed to be built by local administrations and are likely to be occupied in the year (before the next FILOCOM waves from which we compute vacancy rates).

Land observation (BRGM and IGN). We gather land and soil data to compute two proxies of sprawling capacities across municipalities. The first source is a national digital elevation model of 25 metres resolution (BD ALTI, <https://geoservices.ign.fr/bdalti>) obtained from the French institute for geographical analysis (IGN). The second source comes from the *géorisque* website (<https://www.georisques.gouv.fr/>), which is a platform about the risks of natural disasters maintained by the French geological institute (BRGM). The first source is in raster format that we aggregated at the municipal level with a geographical information system. The second biophysical source is readily available at the municipal level to be matched with other variables.

Official statistics (INSEE). In addition to the control variables about dwelling and household characteristics available in FILOCOM,⁷ we use the harmonised population census provided by INSEE over the considered period. These data contain the official population of municipalities used to compute our measure of urban sprawl, and some socio-economic characteristics that could be incidentally correlated with the physical characteristics of the land and confound the

⁷ The computation of control variables from FILOCOM differs from the computation of alternative vacancy rates where the auxiliary variables are interacted with vacancy before their aggregation at the municipal level.

estimates (Newman et al., 2018; Segú, 2020). We also use other INSEE sources about a classification of municipalities along the urban-rural gradient (defined from the continuity of developed areas and daily commuting patterns in 2010, *Zonage en aires urbaines*, Baumont and Guillain, 2016) and a 100 m raster of population density from confidential data.

2.2 Variable computations

Vacancy rates. For all French housing units and each year, FILOCOM report the vacancy status on 1st January according to the occupation declared by the renters or the owners for the coming year. The reliability of this variable is likely to be high because it determines eligibility for the housing and land tax, from which vacant properties are exempt (although certain municipalities have set a vacancy tax).⁸ We calculate the vacancy rates by dividing the number of vacant dwellings by the total number of dwellings in each municipality. The availability of auxiliary variables at the housing scale allows us to compute alternative outcome variables about vacancy. We distinguish vacancy according to housing type (flats and houses), duration (short-term vacancy is considered as less than two years), and the quartiles of housing age before their aggregation in vacancy rates at the municipal level.

Urban sprawl. From the approved residential building permits reported in SITADEL, we compute for each municipality i and each FILOCOM year t the developed land area for residential purposes ΔU_{it} . By noting POP_{it} the official municipality population from INSEE data, we compute our urban sprawl measure with:

$$urban_sprawl_{it} = \Delta U_{it} / POP_{it}. \quad (1)$$

This formula considers the two main dimensions of urban sprawl (land use and population) in a dynamic extension of static measures usually used for France (e.g. Charmes, 2013). This measure also differs from measures based solely on urban footprint (Burchfield et al., 2006) or human mobility (Pendall, 1999) by providing a mix between both. We are aware that no single

⁸ Article 232 of the Tax Code (https://www.legifrance.gouv.fr/codes/article_lc/LEGIARTI000034110718/).

measure of sprawl is universally accepted and that our results are conditioned by this choice, we believe that it nevertheless allows a better understanding of the phenomena under study.

Sprawling capacity. The variations of urban sprawl used to estimate its causal effect on residential vacancy are predicted from the average land slope of the IGN data and the proportion of dwellings at low risk of land instability of the BGRM data. The effect of slope on construction costs is well known since [Saiz \(2010\)](#), while the identification requires control variables against the spurious correlation between slope and unobserved "attractiveness". The use of soil stability (*Retrait Gonflement des Argiles*, RGA) is more original, although it has also been previously used by [Ay et al. \(2018\)](#) on SITADEL building permits. A low RGA risk is known to significantly reduce construction costs, with RGA-related claims under the French natural disaster insurance scheme costing around €7 billion between 1988 and 2011 ([Salagnac, 2015](#)).

Age of Dwellings. This variable is constructed by calculating the difference between the year the building was constructed and the year the data was collected. The results are then aggregated at the municipal level to obtain the mean age of dwellings. We expect that the age of dwellings is pre-determined (hence exogenous) and has a positive effect on the vacancy rate, as the quality of residential service generally decreases with the age of the dwellings.

Comfort of Housing. This variable is about the presence a basic housing requirements. Following INSEE's classification, three levels of comfort are proposed: (i) Without comfort (neither a bathtub nor a shower, or a toilet), (ii) Full comfort (with a bathtub or shower, toilet, and central heating) and (iii) Partial comfort (all other possibilities). We calculate the proportion of each category per commune. Higher quality properties (i.e., those with full comfort) are expected to reduce the vacancy rate because they are more likely to find tenants or buyers. Therefore, the comfort variable is expected to have a negative effect on vacancy rates.

Type of Dwelling. This variable distinguishes individual houses from apartments in collective buildings. According to the French terminology, a collective building is defined as a property designed to accommodate at least two households in two separate dwellings. We compute the

rate of individual houses that represents the percentage of individual houses in the municipality. Considering that urban sprawl is almost always a supply of individual houses, its effect on housing vacancy would depend on the substitution patterns between flats and houses.⁹

Age of Owners. This variable is calculated by determining the difference between the owner's year of birth and the year of data collection from housing tax. This variable pertains exclusively to properties owned by individuals (natural persons). The average age of owners at the municipal level was calculated and employed in the subsequent analysis. It is anticipated that the age of the owner may influence the vacancy rate, given that older owners may be less active in managing and renting out their properties.

Homeownership rate. This variable captures the proportion of dwellings in a municipality that are occupied by their owners. The occupation status of primary residences is classified into five categories based on the combination of the occupation code and the owner type code: P (owner-occupiers); LH (tenants in public housing); LS (tenants in public sector housing like public authorities); LP (tenants in the private sector such as individuals, other legal entities, partners, co-owners as owning a common property); Others (occupants at no cost, rural leases, etc.). We then calculate the proportion of owner-occupiers in each commune. A higher proportion of owner-occupied dwellings may be associated with lower vacancy rates, as owners are typically more invested in maintaining and occupying their properties.

Owners' Location. The location of the owner, whether they live in the municipality or elsewhere, can affect how actively they manage their properties. The FILOCOM database provides information on the owner's residence location with the following categories: C(Same commune); D(Same department); R(Same region); F(Rest of France); E(Foreign countries). We calculate the proportion of owners who do not live in the same department as their property for each commune. It is possible that owners living at a greater distance may find it more challenging to manage their properties, which could potentially lead to an increase in vacancy rates.

⁹ If they are not substitute, urban sprawl is expected to increase vacancy more on houses, whereas apartments can be more impacted with a higher elasticity of substitution.

Social Housing. We also observe the effect that the presence of social housing in a municipality might have on the vacancy rate. For this purpose, we use the share of social housing in the total housing stock of each municipality. This variable was obtained by aggregating data on the legal status of housing from the FILOCOM file. The legal status of owner is provided by the DGFIP in a codification of 10 classes. Using this information, we calculate the proportion of social housing in each municipality.

Income. The average net taxable income per household per year includes the sum of the net taxable incomes of all the occupants of the dwelling from the income declaration, after all authorized deductions. It includes salaries as well as other taxable incomes of all the occupants of the dwelling and those attached to the dwelling. This information is aggregated at the municipality level by averaging at the housing level.

Household size. This variable is calculated from the family situation of the reference person, increased by the number of dependent persons. It includes attached persons or spouses (as per the tax household assignment). We calculate the average household size for each commune using this data from the FILOCOM database.

Employement rate. We measure the economic situation of each municipality by gathering the employment rate instead of the unemployment rate. This approach is supported by well-known evidence demonstrating that the employment rate is less susceptible to seasonal shocks and market fluctuations than the unemployment rate.

Type of municipality. The classification of municipalities is based on the *Zonage en aires urbaines* from INSEE. The latest definition from 2010 divides France into four main types of areas and nine sub-areas. Large Urban Areas (includes large urban centres, their surrounding commuter areas, and multipolarised communes within large urban areas.) ; Other Urban Areas (includes medium-sized centres, their surrounding commuter areas, small centres, and their surrounding commuter areas.) ; Other Multipolarised municipalities ; Isolated municipalities

Outside the Influence of Centres. This classification lets us know if a municipality is urban, peri-urban, or rural. Urban sprawl is expected to be higher far from the urban areas, while the reverse is expected for vacancy rates.

Population density. This variable allows us to account for the distribution of the population in a detailed way. The municipal density grid is based on the distribution of the population within the municipality by dividing the area into 1 km² squares (INSEE 2019).¹⁰ This grid distinguishes urban (dense and intermediate density) and communes (low density and very low density) municipalities, and provides a morphological approach to the territory, which helps capture the impact of population density on vacancy rates.

Population growth. The rate of population growth within a municipality is a crucial variable. Municipalities with growing populations may have lower vacancy rates due to increased housing demand, while those with declining populations may face higher vacancy rates. We use the harmonised population census (SAPHIR) provided by INSEE over the considered period.

Sample definition. The data used in this study covers the odd years from 1995 to 2019, including the year 2010 for administrative reason (our results do not depend on the inclusion of 2010). This period was chosen primarily due to the coverage of the tax data, which begins in 1995 and is collected every two years. The sample is a unbalanced panel of approximately 35,000 French municipalities, with the number of municipalities varying from year to year due to administrative changes and data availability. Furthermore, over the years, new variables have been added to the database, resulting in some variables being unavailable for all years. For instance, information such as the age of dwellings and the duration of occupancy or vacancy of dwellings is not available for the year 1995.

¹⁰ <https://www.insee.fr/fr/information/2114627>

Table 1: Descriptive Statistics

<i>Variables</i>	<i>Description</i>	<i>mean</i>	<i>sd</i>	<i>p50</i>
<i>vacancy rates</i>	The percentage of vacant dwellings in a municipality (Filocom)	9.10	4.89	8.22
<i>vacancy rates in H</i>	Vacant rate of individual house (Filocom)	8.12	4.69	7.19
<i>vacancy rates in F</i>	Vacant rate of flat (Filocom)	17.62	19.11	13.21
<i>duration</i>	The average duration of the vacancy in years (Filocom)	3.48	2.16	3.07
<i>vacancy rates LT</i>	Rate for dwellings vacant for more than 2 years (Filocom)	5.14	4.01	4.16
<i>urban sprawl</i>	Average sprawl by municipality per year in m^2/hbt (Filocom & IN-SEE)	124.47	1050.43	26.61
<i>slope</i>	The mean slope (IGN)	3.84	4.49	2.21
<i>rga low hazard</i>	Proportion of low-hazard soils in the municipality (BRGM)	27.22	43.91	
<i>growth of dwellings</i>	Growth of dwelling's stock (Filocom)	2.28	6.59	1.46
<i>number of permits</i>	The average number of building permits granted by each municipality for new construction per year (Sitadel)	12.40	27.70	5
<i>area of permits</i>	The average land area concerned by building permits in ha (Sitadel)	7.71	38.38	1.64
<i>age of dwellings</i>	The average age of dwellings (Filocom)	105.22	42.74	102.13
<i>without Comfort</i>	Proportion of dwellings without comfort (Filocom)	6.72	7.04	4.29
<i>partial Comfort</i>	Proportion of dwellings with partial comfort (Filocom)	21.39	13.83	18.93
<i>full Comfort</i>	Proportion of dwellings with full comfort (Filocom)	71.89	17.92	75.62
<i>living area</i>	Average living area in m^2 (Filocom)	79.41	16.51	78.41
<i>number of rooms</i>	Number of rooms (Filocom)	3.49	0.56	3.51
<i>house</i>	Proportion of individual house (Filocom)	53.41	32.97	53.73
<i>flat</i>	Proportion of flat per municipality (Filocom)	46.59	32.97	46.27
<i>age of owner</i>	The average age of owner (Filocom)	75.75	72.13	60.23
<i>owner occupied rate</i>	Proportion of owner who live in their own property (Filocom)	54.68	18.07	55.20
<i>non-resid owner</i>	Proportion of owner who reside out of the departement (Filocom)	19.39	13.77	16.22
<i>social housing</i>	Proportion of social dwellings of total units (Filocom)	13.54	12.38	10.62
<i>income</i>	Logarithm of average net taxable income per household per year (Filocom)	10.57	0.65	10.36
<i>household size</i>	Average household size in the municipality (Filocom)	2.05	0.48	2.10
<i>population</i>	Total population in the municipality (INSEE)	1680	8688.21	408
<i>population growth</i>	Population growth in the municipality (INSEE)	1.89	6.00	1.95
<i>employment rate</i>	Rate of employment growth (INSEE)	12.95	89.44	-.29

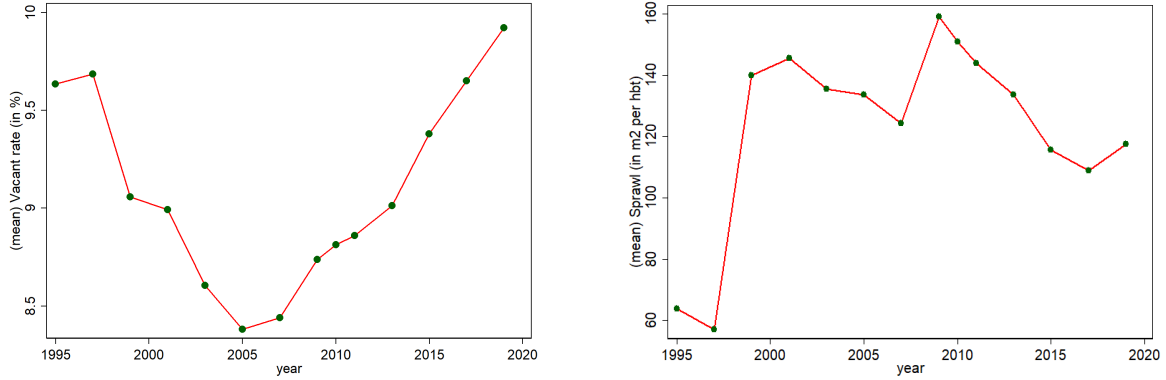
Note : This table presents the summary statistics for the various variables studied in our analysis, covering the period from 1995 to 2019. The variables include vacancy rates, urban sprawl, the number of building permits, housing characteristics, and various socio-economic indicators. Each variable is summarized by its mean, p50 and standard deviation (sd), allowing us to understand their distribution within the data sample. It is important to note that the number of observations is 397,473. This number of observation is due to the incomplete availability of certain variable in some years, particularly in the earlier years of the study. This may be because the database, derived from survey data, has been enriched over time.

2.3 Correlation between sprawl and vacancy

Over the considered period 1995–2019, we obtain an average vacancy rate of 9.10% (Table 1). The dynamics presented in the left panel of Figure 2 are U-shaped: the average residential vacancy rate decreases from 1995 to 2005 and then increases until 2019. This pattern is observed whatever the type of vacancy (short-run or long-run, houses or flats, old or recent housings) computed as alternative outcomes. The first decreasing period could be explained by a favor-

able housing conjuncture with a relatively stable economic growth and low inflation. The trend reversal shortly precedes the financial crisis of 2008, which causes a slowdown of economic growth the probably explain the increase of residential vacancy in the following years.

Figure 2: Trend in residential vacancy rate and urban sprawl from 1995 to 2019



The median urban sprawl per municipality between two FILOCOM waves (about two years) is 27 m² per inhabitant, with an average value of 125 m².inhab⁻¹ (Table 1).¹¹ As in most countries, urban sprawl is positive over the whole considered period (right panel Figure 2). Urban sprawl presents a high increase in 1999 which persists in the following years despite small decreasing trends prior to 2008 and prior to the recent period.¹² The factors from the economic conjuncture that explain these dynamics are rather similar that those that explain residential vacancies, although no clear relationship is apparent between the panels of Figure 2. It is important to note that the identification we propose does not rely on these time variations because we pool the data, we use instruments constant over time, and we include year fixed effects.

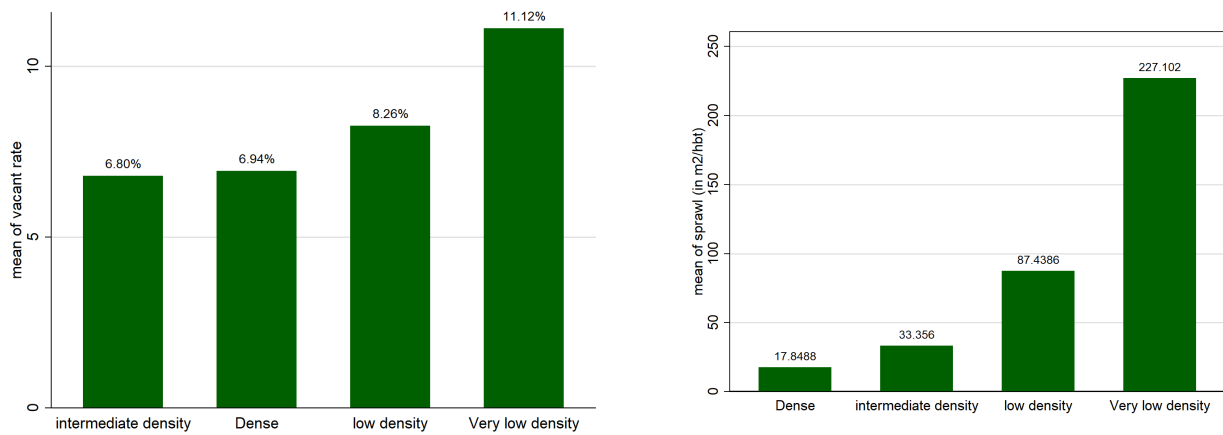
The cross-sectional variations of residential vacancy and urban sprawl between municipalities are illustrated by crossing them with four density categories from the fine INSEE data about population densities (Figure 3). We observe that these two variables decrease with population density with, as expected, a stronger effect for urban sprawl. Although we observe an average vacancy rates higher than about 5 percentage points in very low density municipality (municipalities where more than 50% of the population lives outside any grid are "very low density".)

¹¹ The average land area concerned by building permits is about 7,700 m² for a median to about 1,600 m² (Table 1). The top 1% of the largest building permits accounts for 47% of land development.

¹² The 1999 shift must not be over-interpreted because it is amplified by the increase of data reliability.

compared to municipalities with intermediate density (if more than 50% live in an urban centre or urban cluster, the municipality is "intermediate density"), the raw correlation coefficient between these variables is not significantly different from zero with a value equals to 0.0315. A unreported analysis gives that this correlation with density is also present within French administrative regions. It is worth noting that the absence of the raw correlation between sprawl and vacancy is biased toward zero by the "attractiveness" problem presented in the introduction. Hence, this descriptive statistic does not rule out the existence of undesirable effects on residential vacancy as an unexpected consequence of urban sprawl.

Figure 3: Vacant rate and Sprawl according to density grid



We also find that vacancy rates vary significantly with the age of dwellings with a difference of 5 percentage points between dwellings aged over 100 years and dwellings under 50 years (that present vacancy rates of respectively 10.5% and 5.3%)¹³. This observation can be attributed to the deteriorated condition of older housing units, jointly with the decrease of the quality of the associated residential amenities. [Table 1](#) indicates that average vacancy rate for individual houses is 8.12% while it is 17.62% for flats. Furthermore, our sample indicates an average housing age of about 105 years, emphasising the potential decrease in state of the French housing stock, which is nevertheless more spacious, with an average of around 4 rooms per dwelling. The average age of homeowners in our study is 76 years, and each municipality has an estimated average of 1680 inhabitants, with minimal population growth, reflected in an median population increase of just 1.95 %. Additionally, the average household size in the municipalities is 2.05

¹³ Refer to [Table 16](#) in appendix

persons, which is typical for urban areas. As for average net taxable income per household, it is €52,690.39 per year, with a notable standard deviation of €45,023.11, which shows significant income disparities. Owner-occupied dwellings make up 54.68% of the housing stock, reflecting a balanced mix of homeownership and rental properties. Notably, 16.22% of property owners live outside the department where their property is located, suggesting a level of absentee ownership that may impact property maintenance and vacancy rates. Moreover, 71.89% of dwellings are fully equipped with amenities like bathtubs, showers, and central heating.

3 Empirical framework

3.1 Econometric specification

As shown previously, the regression of residential vacancy on urban sprawl from ordinary least squares (OLS) does not take into account the potential endogeneity bias coming from unobserved "attractiveness" attributes of municipalities. From the standard omitted variables bias formula (Wooldridge, 2010), we expect that OLS under-estimates the true causal effect of urban sprawl because "attractiveness" is expected to drive positively urban sprawl and negatively residential vacancy. Hence, we propose an instrumental variable (IV) estimation that is more usefully described by two-stage least squares procedure.¹⁴

The first step consists in regressing $urban_sprawl_{it}$ defined in Equation 1 for each municipality $i = 1, \dots, N$ and each FILOCOM years $t = 1995, 1997, \dots, 2019$ as follows:

$$urban_sprawl_{it} = \alpha_0 + \alpha_1 Slope_i + \alpha_2 RGA_i + \sum_{k=1}^K \gamma_k X_{kit} + v_t + \xi_{it} \quad (2)$$

The unknown coefficients α , γ , and v are respectively associated to the time-constant $Slope_i$ and RGA_i instruments, a set of K control variables X_{kit} (about the average housing, household, and location of municipalities), and year indicators. As any first stage of an IV estimation, the errors ϵ_{it} are orthogonal to explanatory variable by definition (Wooldridge, 2010). The

¹⁴ The estimations are conducted with the stata command `ivregress 2sls`, accounting in the second stage that the endogenous explanatory variable is predicted.

causal impact of urban sprawl on residential vacancy rates is then obtained in a second stage regression, which consists in regressing $vacancy_rate_{it}$ on the predictions $\widehat{urban_sprawl}_{it}$ obtained from Equation 2 by using the values of all predictors (instruments and controls):

$$vacancy_rate_{it} = \beta_0 + \beta_1 \widehat{urban_sprawl}_{it} + \sum_{k=1}^K \theta_k X_{kit} + \mu_t + \varepsilon_{it} \quad (3)$$

Under the identification restrictions presented in the next subsection, the unknown coefficient β_1 gives the causal effect of increasing urban sprawl by one unit in a random municipality on its vacancy rates (in percentage points). As it is standard for the IV method, the control variables used to lighten the identification restrictions are required to be the same in both stages. The associated coefficients are allowed to change between equations (they are noted θ_k in the second stage instead of γ_k in the first stage). The OLS estimation of Equation 3 is our preferred empirical specification to obtain the causal effect of urban sprawl on residential vacancy. However, we also use alternative outcomes by distinguishing residential vacancy in terms of housing type (flat or house), duration (more or less than two years), and the quartiles of housing age estimated for the same first stage regression.

3.2 Identification restrictions

The IV estimation method relies on some critical identification restrictions that ensure the validity and reliability of the results. These restrictions are typically evaluated through empirical tests and theoretical intuitions. The key conditions are as follows:

1. **Relevance Condition:** The instruments $Slope_i, RGA_i$ must have a strong predictive relationship with the variable $urban_sprawl_{it}$. This requirement implies that there is a non-zero covariance between the two sets of variables, it is usually assessed using the F-test for joint significance of the coefficients α_1 and α_2 in the first-stage regression. This relevance condition holds if the F-statistic for the test that is significantly high, typically above 10. If this F-test indicates that the instruments are weak (i.e., the instruments do not have significant predictive power for the endogenous variables), the IV estimates will be biased and inconsistent. Hence, rejecting the null hypothesis of weak instruments is essential for the validity of the IV approach

2. **Exogeneity Condition:** The instrument $Slope_i, RGA_i$ must be exogenous, meaning it should not be correlated with the error term ε_{it} in the second-stage regression. This condition ensures that the instrument affects the dependent variable only through its impact on the endogenous regressor, and not through any omitted variables or measurement errors. This is often referred to as the exclusion restriction. If one of the instrument is correlated with the errors of the second stage, the IV approach is invalid and the estimates are biased.

In summary, the validity of IV estimation hinges on these critical conditions: the relevance of the instruments, their exogeneity, and the correct exclusion from the structural model. Empirical tests such as the F-test for instrument strength and theoretical justifications for the exclusion restriction are necessary steps in verifying these conditions.

3.3 Over-identification test

The empirical model have two instruments for one endogenous explanatory variable, so the number of instruments is greater than the number of endogenous variables to be instrumented, and the model is said to be over-identified. This makes it possible to use Sargan tests to test the joint exogeneity of the instruments. The IV method is accompanied by two tests of instrument validity, namely the over-identification restriction test and the instrument weakness test.

The over-identification test (Sargan, or Hansen) determines whether the model is over-identified, meaning there are enough instruments. The underlying assumption is that exogenous instruments should not be correlated with the residual model. Specifically, the Sargan test examines the null hypothesis that the instruments are exogenous instruments, i.e., uncorrelated with the error term, and correctly excluded from the estimated equation. The test statistic is computed as $J = nR^2$ where n is the sample size and R^2 is the coefficient of determination from the regression of the IV residuals on the instruments. Under the null hypothesis of instrument validity, this statistic is asymptotically distributed as a χ^2 distribution with degrees of freedom equal to the number of over-identifying restrictions, i.e., the number of instruments minus the number of endogenous regressors.

The instrument weakness test evaluates the joint significance of the instruments using the F-statistic. This test assesses the null hypothesis that the instruments lack strength, meaning they do not have significant predictive power on the endogenous variables. If the null hypothesis is rejected, it indicates that the instruments are robust and applicable for mitigating endogenous regression bias.

4 Results

4.1 Causal effect of urban sprawl on vacancy

The three odd columns of [Table 2](#) report the first-stage coefficients used to instrument the effect of urban sprawl on vacancy rates of all dwellings, houses, and flats (reported in columns 2, 4, and 6, respectively). As expected, the slope and RGA hazard have a statistically significant effect on urban sprawl with intuitive signs. Slope has a negative effect on urban sprawl, indicating that municipalities with steeper slopes tend to have less urban sprawl, and the proportion of land with a low risk of movement (low RGA hazard) has a positive effect on urban sprawl, indicating that land stability is a constrain to urban sprawl in the average municipality. The bottom panel of the Table shows that the slope and RGA hazard instruments are valid according to the weak instrument F statistics (higher than 10) and the Sargan statistics (unsignificant).

In the second stages, the coefficients associated to urban sprawl are positive and significant whatever the vacancy measure. Specifically, in the second column about the total residential vacancy rate, a median increase of + 26.62 m²/inhab. of the urban sprawl leads to an increase in the vacancy rate of 0.33 points (0.0129×26.62), which represents approximately 2.85 housings by municipality.¹⁵ Because households' preferences are likely to be different between flats and detached houses, we examine how urban sprawl might affect vacancy rates for each type of dwelling. The results in columns 4 and 6 of [Table 2](#) show that urban sprawl has a positive effect on both types of dwellings, but with a different magnitude. The coefficient of urban sprawl obtained for apartment vacancy is almost four times higher than that obtained when we use

¹⁵ The French housing stock counts about 30 millions housing units in about 35 thousand municipalities.

house vacancy as the dependent variable. This significant results highlight that, other things equal, households have a preference for detached houses and there is a strong substitution of housing demand from flats to house in the municipalities with urban sprawl.

The coefficients obtained for the age and comfort variables support the hypothesis of a mismatch between the existing housing stock and the preferences of households. Specifically, the coefficient associated with the age of the existing dwellings is positive and significant across all specifications. This indicates that the average age of the housing stock in a municipality positively influences the residential vacancy rate. This phenomenon can be attributed to older dwellings being prone to obsolescence, often requiring extensive and costly renovations that many owners cannot afford. Furthermore, older homes were typically built to meet the standards and needs of a different era and, thus, no longer align with the expectations of modern tenants (Brueckner, 1980). In addition, the coefficient associated to the comfort variable is another significant finding that supports our hypothesis. It is negative and statistically significant, implying that a one-percentage-point increase in the number of dwellings that provide full comfort within a region leads to a decrease in the overall vacancy rate by 0.046 percentage points. This suggests that higher-quality housing units are more likely to meet the expectations of potential occupants, reducing vacancy rates.

Regarding the variables that characterize households present in municipalities, our results reveal a positive effect of the age of the owner on the vacancy rate. This can be primarily attributed to older homeowners transitioning to elderly care facilities or relocating to southern areas, leading to increased vacancies in urban areas. Conversely, the average household size has a negative and significant effect on the vacancy rate. However, when considering the flat vacancy rate as a measure of residential vacancy, the coefficient turns positive. This can be explained by the fact that houses, typically located on the outskirts, offer larger living spaces suitable for large families, whereas flats are usually situated in densely populated urban areas, providing limited space and being less suitable for larger households.

Finally, our findings indicate that the demographic and economic profile of a municipality correlates with its vacancy rate. Specifically, the negative and highly significant coefficient associated with the employment rate suggests that favorable economic conditions attract new

residents, thus reducing the vacancy rate. These observations align with the previous study conducted by [Newman et al. \(2018\)](#) in the United States, which found similar correlations between economic conditions and residential vacancy rates.

Table 2: 2SLS estimates of the impact of urban sprawl on residential vacancy

	<i>Residential vacancy rates for:</i>					
	All dwellings		Only houses		Only flats	
	1st stage	2nd stage	1st stage	2nd stage	1st stage	2nd stage
slope	-1.939*** (0.386)		-1.939*** (0.386)		-1.939*** (0.386)	
rga low hazard	1.941** (0.839)		1.941** (0.839)		1.941** (0.839)	
urban Sprawl		0.0129*** (0.00249)		0.0380*** (0.00712)		0.148*** (0.0333)
age of dwellings	0.361*** (0.0461)	0.0115*** (0.00110)	0.361*** (0.0461)	0.00292 (0.00314)	0.361*** (0.0461)	0.0214** (0.0106)
full comfort	-1.005*** (0.141)	-0.0466*** (0.00289)	-1.005*** (0.141)	-0.0305*** (0.00826)	-1.005*** (0.141)	0.0762** (0.0373)
household size	-68.98*** (7.689)	-2.903*** (0.184)	-68.98*** (7.689)	-0.303 (0.524)	-68.98*** (7.689)	6.970*** (2.400)
income	27.91*** (8.712)	-1.502*** (0.110)	27.91*** (8.712)	-2.340*** (0.347)	27.91*** (8.712)	-4.538*** (1.370)
age of owners	-5.089*** (0.720)	0.157*** (0.0159)	-5.089*** (0.720)	0.259*** (0.0453)	-5.089*** (0.720)	0.804*** (0.212)
social housing	-2.362*** (0.294)	0.133*** (0.00697)	-2.362*** (0.294)	0.106*** (0.0199)	-2.362*** (0.294)	0.283*** (0.0758)
non-resid owners	1.800*** (0.215)	-0.116*** (0.00524)	1.800*** (0.215)	-0.136*** (0.0150)	1.800*** (0.215)	-0.360*** (0.0615)
Employment rate	0.115*** (0.0192)	-0.00251*** (0.000376)	0.115*** (0.0192)	-0.00550*** (0.00107)	0.115*** (0.0192)	-0.0208*** (0.00471)
Peri-urban	19.61*** (6.335)	0.0129 (0.1000)	19.61*** (6.335)	-0.258 (0.285)	19.61*** (6.335)	-0.537 (1.114)
small urban area	10.20 (8.252)	0.670*** (0.113)	10.20 (8.252)	0.307 (0.324)	10.20 (8.252)	1.778 (1.159)
not in urban areas	76.20*** (7.057)	0.239 (0.221)	76.20*** (7.057)	-1.362** (0.630)	76.20*** (7.057)	-6.524** (2.692)
Constant	427.0*** (59.19)	10.76*** (1.256)	427.0*** (59.19)	-1.377 (3.586)	427.0*** (59.19)	-40.30** (17.08)
Observations	397,473	397,473	397,473	397,473	397,473	397,473
Year fixed effects	yes	yes	yes	yes	yes	yes
F-stat	92.04	643.0	92.04	77.05	92.04	88.875
Sargan		0.738		0.121		0.1567

Notes : The table reports the estimated coefficients associated to [Equation 3](#). The dependent variable in first stages is urban sprawl, and the causal impact of sprawl is estimated on residential vacancy in all dwellings, then on residential vacancy in houses only, and finally on residential vacancy in flats in the second stages. The sample period spans from 1995 to 2019, all specifications include year fixed effects. Clustered spatial (at the *municipality* level) and time standard errors in brackets. ***p < 0.01, **p < 0.05, *p < 0.1. For detailed variable descriptions and sources see [Table 1](#).

In [Table 3](#), we re-estimate our model using the structural vacancy rate (the proportion of dwellings vacant for more than two years) as the dependent variable. This measure is used

to test whether urban sprawl has a permanent effect on residential vacancy or if this effect is mainly related to temporary vacancy. The instruments (slope and RGA risk) consistently show statistically significant effects on urban sprawl. It should be noted that all specifications in this table are validated both by the Sargan over-identification test and the F-test. The results confirm that urban sprawl has a positive and significant effect on structural vacancy. Specifically, an increase in urban sprawl by 25 m²/inhab. leads to an increase of 0.17 percentage points in the overall structural vacancy rate, which represents approximately 1.5 housings by municipalities. This result remains consistent whether we are discussing structural vacancy for flats or houses. Nevertheless, the coefficients for urban sprawl on total residential vacancy ([Table 2](#)) are nearly twice as high as those for structural vacancy ([Table 3](#)) for all types of dwellings. Although the effect of urban sprawl is relatively smaller on structural vacancy, it remains significant and poses a substantial problem for housing affordability.

Table 3: 2SLS estimates with alternative measure of vacancy : Structural vacancy rate

	<i>Structural Residential vacancy rates for:</i>					
	All dwellings		Only houses		Only flats	
	1st stage	2nd stage	1st stage	2nd stage	1st stage	2nd stage
slope mean	-1.939*** (0.386)		-1.961*** (0.386)		-1.133*** (0.346)	
rga low hazard	1.962** (0.839)		1.941** (0.838)		2.635** (.762)	
urban Sprawl		0.00670*** (0.00136)		0.0185*** (0.00345)		0.0874*** (0.0195)
age of dwellings	0.361*** (0.0461)	0.0101*** (0.000599)	0.363*** (0.0461)	0.00600*** (0.00153)	0.361*** (0.0461)	-0.00999 (0.00626)
full comfort	-1.005*** (0.141)	-0.0490*** (0.00158)	-1.005*** (0.141)	-0.0376*** (0.00404)	-1.005*** (0.141)	0.00835 (0.0221)
household size	-68.98*** (7.689)	-2.034*** (0.100)	-68.98*** (7.689)	-0.846*** (0.254)	-68.98*** (7.689)	3.687*** (1.401)
income	27.91*** (8.712)	-1.015*** (8.712)	27.91*** (8.712)	-1.436*** (0.0568)	27.91*** (0.165)	-2.264*** (0.815)
age of owners	-5.089*** (0.720)	0.0928*** (0.00865)	-5.089*** (0.720)	0.131*** (0.0219)	-5.089*** (0.720)	0.483*** (0.124)
social housing	-2.362*** (0.294)	0.0478*** (0.00380)	-2.362*** (0.294)	0.0390*** (0.00969)	-2.362*** (0.294)	0.152*** (0.0446)
non-resid owners	1.800*** (0.215)	-0.0777*** (0.00286)	1.800*** (0.215)	-0.0855*** (0.00727)	1.800*** (0.215)	-0.212*** (0.0361)
Employment	0.115*** (0.0192)	-0.00176*** (0.000205)	0.115*** (0.0192)	-0.00311*** (0.000524)	0.115*** (0.0192)	-0.0129*** (0.00278)
Peri-urban	19.61*** (6.335)	0.277*** (0.0545)	19.61*** (6.335)	-0.0579 (0.139)	19.61*** (6.335)	-0.0520 (0.657)
small urban areas	10.20 (8.252)	0.715*** (0.0618)	10.20 (8.252)	0.308* (0.159)	10.20 (8.252)	1.532** (0.688)
not in urban areas	76.20*** (7.057)	0.833*** (0.120)	76.20*** (7.057)	-0.150 (0.305)	76.20*** (7.057)	-3.096** (1.578)
Constant	427.0*** (59.19)	7.644*** (0.685)	423.9399*** (59.22)	2.587 (1.737)	462.18*** (55.83)	-23.46** (9.979)
Observations	397,473	397,473	397,330	397,330	346610	346610
Year fixed effects	yes	yes	yes	yes	yes	yes
F-stats	92.04	14.29	92.04	14.61	92.04	10.37
sargan		0.245		0.945		0.157

Note: The Table presents the estimation results for only those dwellings that have been vacant for at least two years. The dependent variable in the first stages is always urban sprawl. In the second stages, the impact of urban sprawl is estimated firstly on structural residential vacancy rate for all dwellings, then on structural residential vacancy rate for houses only, and finally on structural residential vacancy rate for flats. The sample period spans from 1995 to 2019. All specifications include year fixed effects. Clustered spatial (at the *municipality* level) and time standard errors in brackets. ***p < 0.01, **p < 0.05, *p < 0.1. For detailed variable descriptions and sources see [Table 1](#).

4.2 Robustness checks

The main threat to identification concerns the spatial spillovers across municipalities. For example, the vacancy rate of a municipality with a low urban sprawl could be impacted by the high urban sprawl of its neighbors. To assess the robustness of our findings, we firstly aggregated our data at the job area level (also known as commuting zones). This approach is expected to capture clusters of municipalities with similar labour markets and economic trends, providing a realistic representation of areas where people live and work. Aggregating at this level helps to control for spatial variations and reduces issues related to spatial autocorrelation, ensuring that our analysis reflects broader trends rather than spillovers.

The [Table 4](#) below displays the findings at job areas level. The data establishes the positive impact of urban sprawl, which varies according to our measures of residential vacancy for all dwellings, only houses, and only flats. The strength of this impact is more marked where an increase of + 26.64 m²/inhab. in urban sprawl results in a 0.50 percentage point rise in the overall vacancy rate, and in a 4.64 percentage point rise in the vacancy rate for flats. To recap, our results at the level of job areas underline the positive impact of urban sprawl on residential vacancy rates even more clearly and highlight the importance of local dynamics in this relationship.

Another important threat for identification is the structural change in the housing markets following the 2008 crisis. We divide our sample into two distinct periods, before and after the 2008 financial crisis ([Table 9](#) in appendix). This approach will allow us to analyse how the impact of urban sprawl on residential vacancy may have changed over time, taking into account the crisis-induced fluctuations. The results show that the positive effect of urban sprawl on vacancy is maintained whether we consider the period before or after the 2008 crisis. However, the magnitude of the effect tends to be more pronounced in the post-crisis period ([Table 9](#) in appendix). The findings demonstrates a heightened impact in the aftermath of the economic crisis. This result can be attributed to the perception of real estate as a secure investment during the 2008 recession, resulting in an appreciable rise in property value (+12 % between 2009 and 2011). Consequently, the transaction volume plummeted to 800,000 in 2011, mirroring the pre-crisis level in 2007. Nonetheless, the implementation of more stringent criteria for property

loans has restrained this trend. We must await until 2016 for interest rate reductions to revitalize the property market.

To account for spatial heterogeneity, we exclude the French overseas territories (DOM) from our sample (Table 10 in appendix). This decision was motivated by the significant differences in the demographic, economic and geographical characteristics of the DOM compared with metropolitan France, which could potentially bias our results. Also, we have observed levels of residential vacancy in the DOMs that are significantly different from those in metropolitan France (e.g. a vacancy rate of 30% in Mayotte). Our objective is to evaluate whether our results are biased by the French Overseas Departments by excluding them. The results are presented in the in below. They show that our findings remain robust even after excluding the French Overseas Territories from our analysis. The impact of urban sprawl on residential vacancy remains positive and significant. What's more, we do not observe any major variations in the magnitude of this effect.

Table 4: 2SLS estimates from the aggregation at the scale of job areas (*zone d'emploi*)

	<i>Residential vacancy rates for:</i>					
	All dwellings		Only houses		Only flats	
	1st stage	2nd stage	1st stage	2nd stage	1st stage	2nd stage
slope	-1.162*** (0.100)		-1.162*** (0.100)		-1.162*** (0.100)	
rga low hazard	3.137*** (0.400)		3.137*** (0.400)		3.137*** (0.400)	
urban sprawl		0.019*** (0.005)		0.008** (0.004)		0.176*** (0.014)
Constant	-206.0*** (27.39)	2.337 (2.281)	-206.0*** (27.39)	12.49*** (2.031)	-206.0*** (27.39)	51.97*** (6.877)
Observations	3,731	3,731	3,731	3,731	3,731	
Controls	yes	yes	yes	yes	yes	yes
Year fixed effects	yes	yes	yes	yes	yes	yes
F-Stat	77.00	13.65	77.00	13.65	77.00	13.448
Sargan		0.132		0.795		0.000

Note : This table presents the results of estimates obtained by aggregating data at the scale of job areas. The first and second stages were conducted in accordance with the methodology outlined in Table 2. The sample period spans from 1995 to 2019. All specifications include year fixed effects. Clustered spatial (at the *job areas* level) and time standard errors in brackets. ***p < 0.01, **p < 0.05, *p < 0.1. Main controls are those of Table 2.

4.3 Heterogeneity analysis

This section discusses the heterogeneity of the causal effect of urban sprawl on vacancy to shed some light on the underlying mechanisms. Specifically, we discuss the role of (i) initial vacancy levels, (ii) dwellings' ages, (iii) employment dynamics, (iv) previous control variables, (v) local tensions in the housing market, (vi) population density, and (vii) coastal law to mediate the effect of urban sprawl on vacancy rates.

Subsampling by initial vacancy levels. The impact of urban sprawl can be first examined by distinguishing the quartiles of the residential vacancy rate. Dividing the sample into quartiles according to the residential vacancy rate in 1995 enables the identification of significant differences between areas. This analysis aims to investigate whether zones with high vacancy rates react differently to urban sprawl than those where vacancy is less of matter. To achieve this, we divide our sample into quartiles according to the residential vacancy rate and estimate the model separately. The findings reported in [Table 5](#) indicate that the impact of urban sprawl is more noticeable in regions where the vacancy rate is above the median. Urban sprawl has a greater impact in areas where there are already higher vacant housings. This may be explained by the fact that in places where vacancy rates are high, the addition of new buildings or the development of urban areas may have a cumulative effect on the number of vacant dwellings. High vacancies may also reflect reduced demand for housing. Sprawl can exacerbate this situation by adding more supply to the market without necessarily increasing demand significantly. Some areas appear to be more susceptible to the consequences of urban sprawl than others, according to this result. To address residential vacancy in these areas, local authorities should examine implementing targeted measures.

Subsampling by average age of dwellings. According to the literature ([Brueckner, 1981](#)), the age of a dwelling influences the probability that the dwelling is vacant. The age of the dwelling would thus be a determining factor in the occupancy of a dwelling. This is because older housing is prone to obsolescence and thus requires extensive renovation, which usually proves to be too expensive for homeowners. In addition, housing that is old is for the most part built accord-

Table 5: 2SLS estimates by quartiles based on vacant rate (in %)

<i>Dependent variable:: Residential vacancy rates for all dwellings</i>								
	Q1		Q2		Q3		Q4	
	Stage 1	Stage 2	Stage 1	Stage 2	Stage 1	Stage 2	Stage 1	Stage 2
slope	-4.739*** (0.952)		-1.533** (0.727)		-2.313*** (0.864)		-5.387*** (0.947)	
rga low hazard	1.561 (1.990)		1.253 (1.570)		1.486 (1.026)		3.414** (1.718)	
urban Sprawl		-0.001* (0.000)		0.000 (0.000)		0.017** (0.007)		0.023*** (0.006)
Constant	897.2*** (129.5)	4.407*** (0.251)	94.45 (109.9)	7.317*** (0.0906)	144.2 (104.4)	10.35*** (0.237)	514.2*** (153.9)	21.65*** (0.664)
Observations	103,119	103,119	104,314	104,314	99,872	99,872	91,536	91,536
Controls	yes	yes	yes	yes	yes	yes	yes	yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	yes	Yes	yes
F-stats	34.63	212.2	18.90	151.0	63.71	65.42	45.88	164.0
sargan		0.000		0.980		0.1464		0.0211

Note : This table reports the results of the estimates obtained using a regression by quartiles based on the vacancy rate value. The quartiles are defined with the thresholds of 5.56% between Q1 and Q2, 8.22% between Q2 and Q3, and 11.7% between Q3 and Q4. The dependent variable in the first stage is urban sprawl. In the second stage, the dependent variable is residential vacancy rates for each quartile value. The sample period spans from 1995 to 2019. All specifications include year fixed effects. Main controls are those of [Table 2](#). Clustered spatial (at the *municipality* level) and time standard errors in brackets. ***p < 0.01, **p < 0.05, *p < 0.1. For detailed variable descriptions and sources see [Table 1](#).

ing to the needs of another era and therefore no longer meets the expectations of today's tenants ([Jeon and Kim, 2016](#)). A final reason is given by the "vintage model" ([Brueckner, 1980](#); [Brueckner and Rosenthal, 2009](#)) which assumes that the level of residential services of a modern dwelling, generally more comfortable, is higher than that of an old dwelling, which is more deteriorated. In this part, we evaluate the impact of sprawl on residential vacancy according to the average age of the housing stock of the municipalities. We want to test whether the effect of sprawl on the residential vacancy rate depends on the age of the housing stock. To do this, we subset our sample into four segments using quartiles based on the age of the housing. The results show that urban sprawl has a positive influence on residential vacancy, regardless of the age range of the housing considered. However, it is worth noting that there are significant variations in the magnitude of this impact. In particular, the effect of urban sprawl on residential vacancy increases as the housing stock ages ([Table 6](#)). These findings reinforce the notion that the construction of new homes competes with older ones, highlighting the dominant role of housing

age in occupancy.

Table 6: 2SLS estimates by quartiles based on age of dwellings

<i>Dependent variable:: Residential vacancy rates for all dwellings</i>								
	Q1		Q2		Q3		Q4	
	1st stage	2nd stage	1st stage	2nd stage	1st stage	2nd stage	1st stage	2nd stage
slope	-3.203*** (0.509)		-3.358*** (0.740)		-0.00314 (0.645)		-6.353*** (1.200)	
rga low hazard	1.421 (0.966)		4.229*** (1.578)		3.364** (1.424)		-2.004 (2.842)	
urban Sprawl		0.001*** (0.000)		0.002*** (0.001)		0.011** (0.005)		0.011*** (0.002)
Constant	622.5*** (75.71)	3.786* (2.243)	366.6*** (123.4)	15.34*** (0.559)	180.3* (106.5)	16.42*** (1.533)	473.0*** (182.5)	16.38*** (2.142)
Observations	110,298	110,298	104,579	104,579	97,650	97,650	84,959	84,959
controls	yes	yes	yes	yes	yes	yes	yes	yes
Year fixed effect	yes	yes	yes	yes	yes	yes	yes	yes
F-stats	39.09	440.6	17.70	1364	23.18	219.1	12.61	53.91
sargan		0.988		0.000		0.000		0.211

Note : This table reports the results of the estimates obtained using a regression by quartiles based on the age of dwellings. The quartiles are defined with the thresholds of 76.195 between Q1 and Q2, 102.13 between Q2 and Q3, and 128.87 between Q3 and Q4. The dependent variable in the first stage is urban sprawl. In the second stage, the dependent variable is residential vacancy rates for all dwellings. The sample period spans from 1995 to 2019. All specifications include year fixed effects. Main controls are those of [Table 2](#). Clustered spatial (at the *municipality* level) and time standard errors in brackets. ***p < 0.01, **p < 0.05, *p < 0.1. For detailed variable descriptions and sources see [Table 1](#).

Subsampling by employment dynamics. We want to test if a municipality's residential vacancy rate is closely linked to its socio-economic stability. This study examines the economic conditions of such municipalities, utilizing employment rate as a principal gauge rather than the more volatile unemployment rate. This employment choice is justified by objective data, as the employment rate is less prone to seasonal variations and market instabilities, as stated by [Levine \(2012\)](#); [Kitov and Kitov \(2011\)](#). Furthermore, the employment rate has been widely acknowledged in existing literature as a reliable indicator of economic growth ([Almeida, 2007](#)). To further explore the causal relationship between urban sprawl and residential vacancy rates, we divide our dataset into two distinct sub-samples based on the evolution of employment levels over our study period. The first sub-sample consists of areas that have experienced an increase in their employment rates, whereas the second sub-sample encompasses those where employment has decreased. Our research confirms the preliminary supposition, demonstrated

in columns 2 and 4 of Table 7. Clearly, urban sprawl has a more significant effect on the vacancy rate in municipalities with declining economic activity.

Table 7: Heterogeneity by economic condition: 2SLS estimates

<i>Dependent variable: Residential vacancy rates for all dwellings</i>				
	Increase in employment		Decrease in employment	
	1st stage	2nd stage	1st stage	2nd stage
slope	-1.717*** (0.582)		-2.205*** (0.509)	
rga low hazard	0.950 (1.305)		3.076*** (1.062)	
urban Sprawl		0.0102*** (0.00358)		0.0137*** (0.00285)
Constant	666.2*** (91.93)	8.028*** (2.482)	173.4** (75.51)	15.30*** (1.132)
Observations	196,525	196,525	200,948	200,948
Controls	yes	yes	yes	yes
Year fixed effect	yes	yes	yes	yes
F-stats	45.30	383.6	53.28	354.2
sargan		0.622		0.483

Note : This table reports the results comparing municipalities according to their economic condition. To this end, the sample was divided into two groups: those with increased employment during the study period and those with reduced employment. The dependent variable in the first stage is urban sprawl. In the second stage, the dependent variable is residential vacancy rates for all dwellings. The sample period spans from 1995 to 2019. Main controls are those of Table 2. Clustered spatial (at the *municipality* level) and time standard errors in brackets. ***p < 0.01, **p < 0.05, *p < 0.1.

Interacting control variables. The results of the analysis of heterogeneity by age and density suggest that urban sprawl induces competition between new and old housing. This stimulates the abandonment of older dwellings in favour of new construction on the outskirts. In fact, the construction of new homes on the outskirts of urban centres often brings a number of advantages not found in older homes in central areas, including more affordable prices, greater comfort, state-of-the-art facilities and larger outdoor areas. As a result, interest in older inner-city homes declines, leading to higher vacancy rates for the latter. The trend towards suburban living is being reinforced by the expansion of public transport services, which makes it easier to access outlying areas and increases the attractiveness of new housing stock. To test these hypotheses further, we conducted a cross-analysis of urban sprawl variables with factors such as comfort, household size and localization variables. Examining interaction effects by cross-

referencing urban sprawl with these factors is essential for a more nuanced understanding of the mechanisms underlying the effect of sprawl on residential vacancy.

Table 8: Interaction effects: 2SLS estimates

	<i>Residential vacancy rates for:</i>						<i>Structural vacancy rate for:</i>	
	All dwellings		Only houses		Only flats		All dwellings	
	1st stage	2nd stage	1st stage	2nd stage	1st stage	2nd stage	1st stage	2nd stage
slope	-0.895** (0.4217)		-0.895** (0.4217)		-0.895** (0.4217)		-0.895** (0.4217)	
rga low hazard	1.8751** (0.8387)		1.8751** (0.8387)		1.8751** (0.8387)		1.8751** (0.8387)	
Urban Sprawl		0.2983** (0.1358)		0.2444** (0.1113)		0.5132*** (0.1754)		0.1091** (0.0498)
sprawl*confort		-0.000* (0.0000)		-0.000* (0.0000)		0.0003*** (0.0001)		-0.000** (0.0000)
sprawl*age of dwelling		-0.0002** (0.0001)		-0.0001** (0.0001)		-0.0003*** (0.0001)		-0.0001** (0.0000)
sprawl*household size		0.0234** (0.0106)		0.0191** (0.0087)		0.0475*** (0.0162)		0.0086** (0.0039)
sprawl*Per-iurban		-0.6450** (0.8037)		-1.4458** (0.6587)		-2.8078*** (0.9598)		-1.7648** (0.2945)
sprawl*small urban areas		-1.7296** (0.7877)		-1.4169** (0.6456)		-2.7199*** (0.9298)		-0.6321** (0.2886)
sprawl*not in urban area		-1.8031** (0.8212)		-1.4772** (0.6730)		-2.8841*** (0.9858)		-0.6590** (0.3009)
Observations	397482	397482	397482	397482	397482	397482	397482	346750
Controls	yes	yes	yes	yes	yes	yes	yes	yes
Year fixed effect	yes	yes	yes	yes	yes	yes	yes	yes
F-test	89.99	10.175	89.99	10.175	89.99	9.20	89.99	10.175
Sargan		0.9420		0.9652		0.5459		0.3270

Note : The equation is estimated by considering the main model augmented by the interactive term. Each variable in isolation (without interaction with sprawl) and controls are included but not reported. Clustered spatial (at the *municipality* level) and time standard errors in brackets. ***p < 0.01, **p < 0.05, *p < 0.1.

These selected variables play a crucial role in distinguishing between new and old buildings, offering a more precise perspective on competitiveness and residential preferences in the housing market. The results are outlined in [Table 8](#). The findings indicate that the introduction of crossed variables significantly strengthens the effect of sprawl on residential vacancy. This confirms that the sprawl effect on vacancy is interacted with other characteristics, notably dwelling comfort, household size, age of the housing stock, and location. These results confirm the assumptions that new housing in the suburbs, with its favourable prices, modern comforts, generous spaces and better accessibility thanks to public transport, encourages residents to move away from the city centres. Also, the result obtained with the household size*sprawl interaction variable indicates that urban sprawl leads to an increase in the size of single-family houses and a reduction in population density, which can result in a high ratio of vacant dwellings to the

population (Kendrick, 1979).

We report other heterogeneity analysis (ABC zoning, density, coastal) in the appendix. In Table 11, the heterogeneity analysis according to the "ABC zoning" indicates that the impact of urban sprawl on vacancy rates is contingent upon the prevailing dynamics of the local property market. In areas characterised by high demand (zones A and ABis), the effect is negligible. Conversely, in less intense market conditions, the impact on vacancy rates increases as market tension diminishes. Next, an examination of the conditional effects according to population density reveals that only sprawl in low and very low-density areas leads to a notable increase in residential vacancy. This highlights the competition between new suburban housing and older central city properties, which is exacerbated by the spread of public transport networks (Table 12). Finally, with regard to the coastal law, the analysis exhibits that urban sprawl exerts a positive influence on residential vacancy in both coastal and non-coastal municipalities, with a more pronounced effect observed in coastal areas. This is largely attributable to property speculation, whereby owners maintain properties unoccupied in anticipation of rising prices (Table 13).

5 Conclusion

The intent of this article is to ascertain the causal effects of urban sprawl on residential vacancy rates in France, employing data from French municipalities spanning the period from 1995 to 2019. In order to achieve this objective, we employ an instrumental variable method with slope and soil stability (*retrait gonflement des argiles*) leveraged as exogenous instruments. The results of our analysis indicate that urban sprawl has a substantial and statistically significant positive impact on residential vacancy rates. In particular, an increase of 1 m² per inhabitant in housing construction results in a 0.0129 percentage point increase in the vacancy rate, which is equivalent to approximately 5,192 properties at the national scale.

It is noteworthy that the impact of urban sprawl is more pronounced in the context of vacant apartments than in the case of vacant houses. Moreover, our analysis demonstrated that the impact of urban sprawl persists when structural vacancy is employed as a measure of vacancy. Ad-

ditionally, the impact of urban sprawl is more pronounced in economically disadvantaged areas, which are characterised by a decline in employment. The presence of high initial vacancy rates also accentuates the effect of urban sprawl, with this effect becoming more pronounced above the median vacancy rate. Additionally, this research demonstrates that the impact of urban sprawl is contingent upon the tightness of the local property market. Furthermore, it appears that the influence of urban expansion on vacancy rates tends to intensify in areas with relatively loose property markets. The introduction of cross-tabulated variables indicates that the impact of urban sprawl is amplified, suggesting that factors such as comfort, the age of dwellings, household size, and the location of municipalities condition this impact. It is important to note that the results remain robust when the scale of analysis is changed by aggregating the data at the employment zone level and when the overseas territories are excluded.

The present study thus constitutes a significant contribution to the understanding of the impact of urban sprawl on residential vacancy in France, while emphasising the diversity of effects as a function of different economic, demographic and geographical variables. The findings presented herewith can be of considerable utility in guiding policy in urban planning and property building with a view to addressing the issue of residential vacancy. Furthermore, this study makes a notable contribution to the existing literature on the housing affordability crisis by challenging the prevailing view among economists that an increase in housing construction is the sole solution to this crisis. Our findings indicate that the housing crisis is not solely a consequence of a lack of housing, but rather the inadequate quality of the available housing stock. It would therefore appear that the strategic option of renovating or refurbishing unoccupied housing would be a more practical and sustainable solution. This option also has an environmental benefit, as it limits the carbon footprint associated with the construction of new homes. As highlighted in our study, it is of the utmost importance to gain a deeper understanding of the multifaceted impacts of urban sprawl in order to develop targeted and effective housing policies that not only increase the quantity but also the quality, and accessibility of housing.

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A ONLINE APPENDIX (NOT FOR PUBLICATION)

A.1 Theory about the cause of vacancy

The theoretical models proposed in the literature to explain the mechanisms leading to vacancies in the housing market are the standard market model and the Search and matching theory.

The standard market model

The standard market model treats housing like any other market good. The market is characterized by: pure and perfect competition, information is complete and symmetric, the agents are rational and the price is determined by the equilibrium between supply and demand (Fallis and Smith, 1984; Fritzsche and Vandrei, 2018). Based on this theory, there are two possible situations in which a unit can be vacant. First, units are only left vacant if the market price is equal to or lower than the owners' minimum price. This minimum price is determined by ownership costs or transaction costs. The second way of explaining residential vacancy in this model is through monopolistic power. If an owner in this market has a significant share of the market power, he may voluntarily choose to reduce the number of units he offers in order to increase prices. Thus, the existence of a monopolistic owner may lead to an elevation of the vacancy rate in a locality (Molloy, 2016).

Search and matching theory

The Search and matching theory introduces imperfect information on the housing market in the standard model. According to this theory, agents need time to find each other and make the transactions (Stigler, 1961). Indeed, the owner must have time to study the requests he receives and the buyer or the tenant also needs a lot of time to find the housing that corresponds to these preferences (Guasch and Marshall, 1985). A significant number of researchers have analyzed housing vacancy by applying this model and have suggested hypotheses in which vacancy can exist or even persist. Firstly, it appears from these applications that the more heterogeneous the housing in the market, the more vacancies there are. There are fewer players in the market for each "housing category", which leads to a long search and matching process. Also, if the housing is too heterogeneous, tenants or buyers need much more time and cost to gather information about the properties and quality of the available housing (Guasch and Marshall, 1985; Forgey, Rutherford, and Springer, 1996; Haurin, 1988). Secondly, the level of the costs of holding a home can condition the vacancy rate. Indeed, the motivation of buyers and sellers to search and wait is related to the costs they face in the process. This conclusion is drawn from the work of Miceli (1989). For this scientist, sellers seek to optimize between the sale price and the cost associated with the time the unit remains on the market. The tenant or buyer weighs his or her utility between a quick transaction or looking for more in the hope of finding a better home. Finally, Read (1988) introduces lags between the search processes of homeowners and tenants as an alternative extension of the standard search model and concludes that the length of notice to terminate the rental contract positively influences the vacancy rate. The author justifies this assertion by the fact that tenants start their search process for another dwelling before informing

the current landlord of their intention to move. Only then does the landlord begin the search process to find a new tenant. During this time of search, the apartment is usually vacant.

Synthesis and contribution to this study

The analysis of the literature presented above shows that residential vacancy is a pluralistic phenomenon whose main causes are: aging and inadequate supply, unattractive locations, heterogeneity of housing, low demand and the weight of regulations. Therefore, in conducting our study, we must consider these different aspects.

Aging and inadequate supply : It is commonly agreed that the mismatch between supply and demand is a major factor in the occurrence of vacancies (Brueckner, 1981; Nam, Han, and Lee, 2016; Tivadar and Jayet, 2006) . This is because older housing units are built for the most part according to the needs of another era and therefore no longer respond to the expectations of current tenants (Jeon and Kim, 2016). Another explanation is given by the "vintage model" ((Brueckner, 1980; Brueckner and Rosenthal, 2009) Another explanation is given by the "vintage model" which assumes that the level of residential services of a modern, generally more comfortable dwelling is higher than that of an older, more degraded dwelling.

The last factor of non-adaptation is related to the fact that the supply is concentrated in apartments, whereas the demand is more oriented towards individual housing (NEICUN et al.). The recent health crisis has increased this preference by accentuating the desire of households to have additional spaces outside their homes.

Unattractive locations : The economic and demographic characteristics of the location areas influence the vacancy phenomenon. In other words, favorable economic conditions are likely to attract new residents and thus reduce the vacancy rate (Newman et al., 2018; Bowman and Pagano, 2000). Rural areas, especially those that are isolated, poorly served and with few amenities, are also having difficulty attracting people. City centers are also declining in interest due to aging properties and congestions.

The weight of regulations : Vacancy can be the result of a rational decision by the owner. Indeed, landlords prefer to leave an empty apartment rather than rent to people they don't trust. The problem is not to find a tenant but to find the right one, someone trustworthy and solvent. This distrustful behavior of landlords is especially accentuated in France because of the high protection of tenants (Gabriel and Nothaft, 2001).

The length of the notice period for termination of the contract is also a potential factor in increasing the vacancy rate. This is because tenants begin their search for another unit before informing the current landlord of their intention to move. Only then does the owner begin the search process to find a new tenant. During this search time, the apartment is usually vacant. The notice of termination of the contract is 3 months for unfurnished dwellings in France. This delay is too much for the reasons mentioned above. It is thus stimulating the phenomenon of vacancy.

A.2 Additional results

Table 9: Period before the 2008 crisis Vs Period after the 2008 crisis

<i>Dependent variable: Residential vacancy rates for all dwellings</i>				
	Period after the 2008 crisis		Period before the 2008 crisis	
	1st stage	2nd stage	1st stage	2nd stage
slope	-0.763 (0.472)		-3.429*** (0.643)	
rga low hazard	2.177** (1.000)		1.643 (1.407)	
urban Sprawl		0.0269*** (0.0104)		0.0128*** (0.000472)
Constant	727.3*** (69.02)	2.696 (7.693)	240.6** (97.56)	5.008*** (0.540)
Observations	218,793	218,793	178,680	178,680
Controls	yes	yes	yes	yes
Year fixed effect	yes	yes	yes	yes
F	86.74	158.4	46.29	199
sarganp		0.0542		0.091

Note : This table reports the results in regard to the role of the financial crisis of 2008 in this context. To this end, the sample was divided into two groups: Period after the 2008 crisis Vs Period before the 2008 crisis. The dependent variable in the first stage is urban sprawl. In the second stage, the dependent variable is residential vacancy rates for all dwellings. The sample period spans from 1995 to 2019. Main controls are those of [Table 2](#). Clustered spatial (at the *municipality* level) and time standard errors in brackets. ***p < 0.01, **p < 0.05, *p < 0.1.

Table 10: Exclusion of French overseas territories (DOM)

<i>Dependent variable: Residential vacancy rates for all dwellings</i>		
	1st stage	2nd stage
slope	-1.939*** (0.386)	
rga low hazard	1.941** (0.839)	
urban Sprawl		0.0119*** (0.00249)
Constant	427.0*** (59.19)	10.76*** (1.256)
Observations	395,473	395,473
Controls	yes	yes
Year fixed effect	yes	yes
F	92.04	643.0
sarganp		0.738

Note : This table reports the results of estimates when we exclude the overseas French territories from our sample. The dependent variable in the first stage is urban sprawl. In the second stage, the dependent variable is residential vacancy rates for all dwellings. The sample period spans from 1995 to 2019. Main controls are those of [Table 2](#). Clustered spatial (at the *municipality* level) and time standard errors in brackets. ***p < 0.01, **p < 0.05, *p < 0.1.

Subsampling by ABC zoning: Local tensions in real estate markets. The aim of this section is to investigate how urban sprawl can have a variable effect on vacancy rates, depending on the degree of tension in the local residential property market. The ABC zoning system, which categorises areas based on their level of tension in the property market, will be used to achieve this. Zone A is considered the most stretched, while zone C is the least stretched. Tense areas are defined by high prices and rents, as well as strong demand in relation to supply. This leads to rising prices, high rents, growing territorial inequalities, and a population that is increasingly excluded from the possibility of living in these areas. Additionally, tight areas are subject to strict regulations on rent-setting, reduced notice periods for tenants, tax on vacant homes, and higher council tax on second homes. Our research hypothesis is that the tightness of the local housing market may mitigate the impact of urban sprawl on vacancy rates. Table 11 in appendix presents the variable impact of sprawl on housing vacancy, depending on the tightness of the local property market. The findings indicate that in areas of high demand (zones A and ABis), urban sprawl, as measured by construction activity, has a negligible effect on the vacancy rate of dwellings. In other areas with a less competitive housing market, urban sprawl has a significant positive effect on vacancy rates, meaning that vacancy rates increase with urban sprawl. Furthermore, it can be observed that the impact of urban sprawl on vacancy rates tends to increase the more relaxed an area is, i.e. the less tight the property market.

Table 11: Heterogeneity by ABC zoning: Local tensions in real estate markets

	<i>Residential vacancy rates for all dwellings</i>									
	Zone A		Zone ABis		Zone B1		Zone B2		Zone C	
	1st stage	2nd stage	1st stage	2nd stage	1st stage	2nd stage	1st stage	2nd stage	1st stage	2nd stage
slope	-0.557 (0.576)		-2.098 (1.733)		1.028 (0.677)		-1.215*** (0.466)		-2.282*** (0.781)	
rga low hazard	4.664*** (1.095)		0.409 (0.520)		0.872 (2.128)		1.448 (1.185)		1.695* (0.882)	
urban Sprawl		-0.0136* (0.00758)		-0.0604 (0.0905)		0.0195*** (0.00396)		0.0388*** (0.0122)		0.0832*** (0.0278)
Constant	107.0** (48.81)	24.98*** (1.371)	15.28 (139.8)	16.16* (9.338)	235.8*** (76.89)	1.412 (3.757)	62.71 (75.55)	16.89*** (5.103)	355.1*** (121.6)	12.59*** (1.911)
Observations	8,364	8,364	971	971	18,570	18,570	45,184	45,184	324,393	324,393
Controls	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
year fixed effect	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
F	10.36	65.36	12.017	13.123	17.14	25.39	18.62	13.04	75.90	192.8
sarganp		0.233		0.0939		0.0875		0.127		0.780

Note : Clustered spatial (at the *municipality* level) and time standard errors in brackets. ***p < 0.01, **p < 0.05, *p < 0.1. Main controls are those of Table 2

Subsampling by density grid. In this section we have examined the heterogeneity of the impact of urban sprawl on residential vacancy, based on a classification of areas according to their density into four categories (high, intermediate, low and very low density zones), which allows us to better understand the evolution of this impact according to the density grid. It has been found that only in low-density contexts, specifically low-density and very low-density areas, does sprawl lead to a notable increase in residential vacancy (Table 12). This highlights the role of sprawl in creating an excess of unoccupied dwellings. These findings confirm the hypothesis that the phenomenon of urban sprawl creates a competitive dynamic between newly built and established housing. This competition arises because new housing developments on the outskirts of cities often offer a number of advantages over their older, centrally located counterparts. These advantages can include more affordable pricing structures, the inclusion of modern amenities and the provision of extensive outdoor spaces, all of which contribute to their attractiveness over older, inner-city housing. As a result, there is a noticeable shift in demand away from these older properties, leading to an increase in their vacancy rates. This shift is exacerbated by the expansion of public transport networks, which increase the accessibility and attractiveness of suburban areas, effectively redistributing housing demand.

Table 12: Heterogeneity by density grid: 2sls estimates

<i>Residential vacancy rates for all dwellings</i>								
	Dense		Intermediate density		Low density		Very Low density	
	1st stage	2nd stage	1st stage	2nd stage	1st stage	2nd stage	1st stage	2nd stage
slope	-0.887*** (0.201)		-0.649*** (0.158)		-1.173*** (0.417)		-3.386* (1.751)	
rga low hazard	-0.759*** (0.183)		0.242 (0.338)		1.315*** (0.449)		3.211 (2.079)	
urban Sprawl		-0.0568 (0.218)		-0.122 (0.318)		0.0159*** (0.00349)		0.0261*** (0.00986)
Constant	17.21 (33.38)	26.65*** (2.141)	23.52 (23.92)	17.47*** (3.037)	319.3*** (52.69)	17.28*** (0.782)	247.1 (249.1)	29.35*** (10.04)
Observations	9,699	9,699	39,805	39,805	211,655	211,279	104,049	102,489
Controls	yes	yes	yes	yes	yes	yes	yes	yes
Year fixed effect	yes	yes	yes	yes	yes	yes	yes	yes
F-stats	6.988	42.33	46.45	35.66	68.41	779.2	25.05	13.88
sarganp		0.321		0.606		0		0.422

Note : This table reports the results of estimates of Heterogeneity by density grid. To do this, we use the municipal grid density. This allows us to take into account the distribution of the population in a detailed way. The municipal density grid (Insee 2019) is based on the distribution of the population within the municipality by dividing the area into 1 km squares. This grid distinguishes between urban communes (dense and intermediate density) and rural communes (low density and very low density). The municipal density grid provides a morphological approach to the territory. The dependent variable in the first stage is urban sprawl. In the second stage, the dependent variable is residential vacancy rates for all dwellings. The sample period spans from 1995 to 2019. Main controls are those of Table 2. Clustered spatial (at the *municipality* level) and time standard errors in brackets. ***p < 0.01, **p < 0.05, *p < 0.1.

Subsampling by French coastal law. The discussion of the heterogeneity related to the coastal law is essential in analysing the causal impact of sprawl on residential vacancy. The specific nature and attractiveness of coastal areas influence the demand for property, which may have different repercussions on vacancy rates compared to other regions. To achieve this, the sample is divided into two groups: coastal municipalities and non-coastal municipalities. The results indicate that urban sprawl has a positive impact on residential vacancy in both coastal and non-coastal municipalities, with a greater effect in coastal areas (Table 13). This difference in the impact of urban sprawl on residential vacancy between coastal and non-coastal areas can be explained mainly by property speculation. Property owners, anticipating rising property values and overvalued future demand, may keep their properties vacant in hopes of benefiting from increased prices. This attitude may be reinforced by the sight of new construction, which homeowners see as an indicator of a growing market. For these owners, new construction serves as confirmation that their expectations could translate into substantial financial gains.

Table 13: Heterogeneity by French coastal law

<i>Dependent variable: Residential vacancy rates for all dwellings</i>				
	Coastal municipalities		Non-Coastal municipalities	
	1st stage	2nd stage	1st stage	2nd stage
slope	1.8930 (1.9327)		-2.8179*** (0.7126)	
rga low hazard	4.3530*** (1.6383)		2.2550*** (0.8042)	
urban sprawl		0.0266* (0.0137)		0.0166*** (0.0019)
constant	156.7720 (298.5106)	6.1533 (8.2477)	754.4745*** (107.4493)	5.8383*** (1.4565)
Observations	13468	13468	386094	386094
Controls	yes	yes	yes	yes
Year fixed effect	yes	yes	yes	yes
F-stats	6.20536	14.99	37.8895	441.28
sarganp		0.0073		0.9891

Note : Clustered spatial (at the *municipality* level) and time standard errors in brackets. ***p < 0.01, **p < 0.05, *p < 0.1. Main controls are those of Table 2

Table 14: Statistics on Permits Granted for New Construction in Each Region in 2019

Region	Overall		Projects with Area greater than average			The top 1% projects		
	n_permits	area(km2)	n_permits	area(km2)	Proportion	n_permits	area (km2)	Proportion
GUADELOUPE	2593	19,55	562	16,70	85,44%	27	5,65	28,90%
MARTINIQUE	1777	14,31	291	12,32	86,14%	27	7,62	53,28%
GUYANE	610	20,19	141	19,61	97,10%	20	17,45	86,43%
LA REUNION	5515	24,56	494	19,96	81,29%	42	10,09	41,08%
MAYOTTE	682	2,57	69	2,20	85,58%	6	0,77	29,77%
ILE DE FRANCE	38451	164,44	2978	138,70	84,35%	314	87,65	53,30%
CENTRE VAL DE LOIRE	18044	124,74	2333	108,70	87,14%	288	68,23	54,69%
BOURGOGNE FRANCHE COMTE	25130	145,70	3327	121,60	83,46%	328	62,75	43,07%
NORMANDIE	26388	173,40	3894	150,90	87,02%	364	77,72	44,82%
HAUTS DE FRANCE	29952	159,24	3066	135,80	85,28%	289	81,71	51,31%
GRAND EST	40006	190,10	3394	156,50	82,33%	355	96,20	50,60%
PAYS DE LA LOIRE	42747	201,96	4420	165,20	81,80%	436	80,86	40,04%
BRETAGNE	45631	200,44	4672	162,10	80,87%	417	76,12	37,98%
NOUVELLE AQUITAINE	66603	358,19	7935	294,50	82,22%	650	155,90	43,52%
OCCITANIE	55522	322,70	6766	271,80	84,23%	572	154,00	47,72%
AUVERGNE RHONE ALPES	73051	414,22	7812	344,90	83,26%	586	217,90	52,60%
PROVENCE ALPES COTE D AZUR	28607	180,96	3449	150,80	83,33%	282	98,54	54,45%
CORSE	3240	18,63	475	15,77	84,65%	43	8,27	44,42%
Total		2735,91		2288,07	83,63%		1307,42	47,79%

Table 15: Statistics on the area of Permits Granted

Permits for new construction		Permits for building on already existing buildings	
	Area per project (m²)		Area per project (m²)
mean	5422.48	mean	4150.775
Q1	349	Q1	253
Q2	780	Q2	650
Q3	1892	Q3	1500
90e	6293	90e	4355
95e	16015	95e	10389
99e	80525	99e	59438
Std. Dev	51684.25	Std. Dev	44612.6
total	504549		415684

Table 16: Vacancy rates by age of dwellings over the period 1995-2019

	Dwellings aged over 100 years	Dwellings aged between 100 and 50 years	Dwellings aged less than 50 years
% of total stock	55.77%	38.97%	5.26%
Vacant rate (in %)	10.51%	7.57%	5.33%

A.3 Department-level new construction statistics

Table 17: Number of permits granted and their surface area for new construction

dep	n_permiss_per_dep	superficie_m2	superficie_km2
1	9462	31205497	31.2055
2	3538	19948121	19.94812
3	2933	29110454	29.11045
4	1616	14698155	14.69816
5	1460	15045648	15.04565
6	4146	18078341	18.07834

Continued on next page

Table 17: Number of permits granted and their surface area for new construction (continued)

dep	n_permis_per_dep	superficie_m2	superficie_km2
7	4828	15678504	15.6785
8	1008	7060326	7.060326
9	1571	34140000	34.14
10	2259	10438292	10.43829
11	3336	12147908	12.14791
12	2955	28192485	28.19249
13	4026	42274097	42.2741
14	6604	47410531	47.41053
15	1826	15950501	15.9505
16	4229	22645039	22.64504
17	10125	26854192	26.85419
18	2126	19731199	19.7312
19	2368	13546878	13.54688
20	10	19482	0.019482
21	4448	20131978	20.13198
22	10292	42620302	42.6203
23	1061	10532426	10.53243
24	4536	31831107	31.83111
25	5117	22362868	22.36287
26	5937	28626623	28.62662
27	5066	26177023	26.17702
28	2702	16941903	16.9419
29	12484	43274192	43.27419
02A	1563	6504203	6.504203
02B	1667	12104309	12.10431
30	7569	18879600	18.8796
31	11663	73236104	73.23611
32	2042	26451988	26.45199
33	18418	88414950	88.41495
34	9482	32539791	32.53979
35	10767	54586886	54.58689
36	1386	16186865	16.18686
37	4751	23348768	23.34877
38	9138	38560110	38.56011
39	2838	18141909	18.14191
40	4366	33733596	33.7336
41	2558	14664697	14.6647
42	6084	26025870	26.02587
43	2324	8261324	8.261324
44	14255	55457666	55.45766

Continued on next page

Table 17: Number of permits granted and their surface area for new construction (continued)

dep	n_permis_per_dep	superficie_m2	superficie_km2
45	4521	33869167	33.86917
46	2200	12395307	12.39531
47	3058	21229399	21.2294
48	1163	5346664	5.346664
49	6245	36880698	36.8807
50	4660	31868237	31.86824
51	3746	23748781	23.74878
52	1441	13623338	13.62334
53	2150	20254889	20.25489
54	4737	19527420	19.52742
55	1580	9030336	9.030336
56	12088	59963283	59.96328
57	6825	31817847	31.81785
58	1202	13377476	13.37748
59	10224	42378642	42.37864
60	4407	27501242	27.50124
61	2618	20631868	20.63187
62	8549	48625787	48.62579
63	5813	34562656	34.56266
64	6629	45375009	45.37501
65	2420	15605055	15.60505
66	4134	15840920	15.84092
67	9004	26653389	26.65339
68	6227	26878313	26.87831
69	11056	59108480	59.10848
70	2858	17260635	17.26064
71	5077	35844997	35.845
72	5114	33417161	33.41716
73	4682	72998494	72.9985
74	8968	54131226	54.13123
75	6626	11605499	11.6055
76	7440	47315223	47.31522
77	7205	40575705	40.57571
78	5649	26878070	26.87807
79	3310	16899392	16.89939
80	3234	20783368	20.78337
81	4137	26635304	26.6353
82	2850	21293341	21.29334
83	11152	61799492	61.79949
84	6207	29063743	29.06374

Continued on next page

Table 17: Number of permits granted and their surface area for new construction (continued)

dep	n_permis_per_dep	superficie_m2	superficie_km2
85	14983	55945636	55.94564
86	4915	26296280	26.29628
87	3588	20834079	20.83408
88	3179	21320393	21.32039
89	2422	13691658	13.69166
90	1168	4885852	4.885852
91	5601	24718002	24.718
92	2903	8691117	8.691117
93	3060	16487483	16.48748
94	2737	19429328	19.42933
95	4670	16056245	16.05625
971	2593	19551574	19.55157
972	1777	14308340	14.30834
973	610	20190153	20.19015
974	5515	24559577	24.55958
976	682	2570663	2.570663